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Master: Data Science

Assignment: Advanced set of SPARQL queries

Create the SPARQL query and the result for the following queries expressed in natural language. The endpoint that you can use for this exercise is: <https://dbpedia.org/sparql>

Note: *es.dbpedia.org was unavailable (503 error), so dbpedia.org/sparql was used.*

1. Get all the properties that can be applied to instances of the Politician class
(<<http://dbpedia.org/ontology/Politician>>)

```
# RDF base vocabulary namespace
prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
prefix dbo: <http://dbpedia.org/ontology/>
select distinct ?prop
where {
  ?politician rdf:type dbo:Politician .
  ?politician ?prop ?val
}
```

Results

prop
http://www.w3.org/1999/02/22-rdf-syntax-ns#type
http://www.w3.org/2002/07/owl#sameAs
http://xmlns.com/foaf/0.1/name
http://www.w3.org/2000/01/rdf-schema#label
http://dbpedia.org/property/honorificSuffix
http://dbpedia.org/ontology/deathPlace
http://dbpedia.org/property/deathPlace
http://dbpedia.org/ontology/deathDate
http://dbpedia.org/ontology/birthPlace
http://dbpedia.org/property/birthPlace
http://dbpedia.org/ontology/birthDate
http://purl.org/dc/terms/subject
http://dbpedia.org/ontology/wikiPageWikiLink
http://dbpedia.org/ontology/wikiPageExternalLink
http://dbpedia.org/ontology/wikiPageRedirects
http://dbpedia.org/ontology/description
http://xmlns.com/foaf/0.1/homepage
http://dbpedia.org/property/name
http://xmlns.com/foaf/0.1/depiction
http://dbpedia.org/property/almaMater
http://dbpedia.org/property/knownFor
http://dbpedia.org/ontology/almaMater
http://dbpedia.org/ontology/birthName

http://dbpedia.org/ontology/country
http://dbpedia.org/ontology/education
http://dbpedia.org/ontology/militaryService
http://dbpedia.org/ontology/nationality
http://dbpedia.org/ontology/occupation
http://dbpedia.org/ontology/originalName
http://dbpedia.org/ontology/parent

Q1: Showing the first 31 of more than 2,000 results.

2. Get all the properties, except `rdf:type`, that can be applied to instances of the Politician class

```
#RDF base vocabulary namespace
prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
prefix dbo: <http://dbpedia.org/ontology/>
select distinct ?prop
where {
  ?politician rdf:type dbo:Politician .
  ?politician ?prop ?val .
  filter(?prop != rdf:type)
}
```

Results

prop
http://www.w3.org/2002/07/owl#sameAs
http://xmlns.com/foaf/0.1/name
http://www.w3.org/2000/01/rdf-schema#label
http://dbpedia.org/property/honorificSuffix
http://dbpedia.org/ontology/deathPlace
http://dbpedia.org/property/deathPlace
http://dbpedia.org/ontology/deathDate
http://dbpedia.org/ontology/birthPlace
http://dbpedia.org/property/birthPlace
http://dbpedia.org/ontology/birthDate
http://purl.org/dc/terms/subject
http://dbpedia.org/ontology/wikiPageWikiLink
http://dbpedia.org/ontology/wikiPageExternalLink
http://dbpedia.org/ontology/wikiPageRedirects
http://dbpedia.org/ontology/description
http://xmlns.com/foaf/0.1/homepage
http://dbpedia.org/property/name
http://xmlns.com/foaf/0.1/depiction
http://dbpedia.org/property/almaMater
http://dbpedia.org/property/knownFor
http://dbpedia.org/ontology/almaMater
http://dbpedia.org/ontology/birthName
http://dbpedia.org/ontology/country
http://dbpedia.org/ontology/education
http://dbpedia.org/ontology/militaryService

http://dbpedia.org/ontology/nationality
http://dbpedia.org/ontology/occupation
http://dbpedia.org/ontology/originalName
http://dbpedia.org/ontology/parent
http://dbpedia.org/ontology/party
http://dbpedia.org/ontology/profession
http://dbpedia.org/ontology/residence
http://dbpedia.org/ontology/termPeriod
http://dbpedia.org/ontology/thumbnail
http://dbpedia.org/property/alongside
http://dbpedia.org/ontology/committee
http://dbpedia.org/property/wikiPageUsesTemplate
http://dbpedia.org/property/birthDate
http://dbpedia.org/property/citizenship
http://dbpedia.org/property/constituency
http://dbpedia.org/property/deathDate
http://dbpedia.org/property/education
http://dbpedia.org/property/nativeNameLang
http://dbpedia.org/property/occupation
http://dbpedia.org/property/office
http://dbpedia.org/property/order

Q2: Showing the first 48 of more than 2,000 results.

- Which different values exist for the properties, except for `rdf:type`, applicable to the instances of `Politician`?

```
#RDF base vocabulary namespace
prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
prefix dbo: <http://dbpedia.org/ontology/>
select distinct ?prop ?val
where{
  ?politician rdf:type dbo:Politician .
  ?politician ?prop ?val .
  filter(?prop != rdf:type)
}
order by ?prop
```

Results

prop	val
http://dbpedia.org/ontology/Person/height	"170.0"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"181.0"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"178.0"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"176.0"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"174.0"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"165.0"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"187.96"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"175.26"^^<http://dbpedia.org/datatype/centimetre>
http://dbpedia.org/ontology/Person/height	"179.0"^^<http://dbpedia.org/datatype/centimetre>

http://dbpedia.org/ontology/Person/height	"185.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"183.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"175.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"167.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"157.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"170.18"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"173.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"193.04"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"163.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"177.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"188.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"185.42"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"180.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"166.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"168.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"127.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"177.8"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"171.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"154.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"172.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"182.88"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"186.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"190.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"169.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"162.56"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"104.14"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"152.4"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"180.34"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"194.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"187.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"0.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"200.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/height	"198.0"^^< http://dbpedia.org/datatype/centimetre >
http://dbpedia.org/ontology/Person/weight	"83.916"^^< http://dbpedia.org/datatype/kilogram >
http://dbpedia.org/ontology/PopulatedPlace/areaTotal	"47.76"^^< http://dbpedia.org/datatype/squareKilometre >
http://dbpedia.org/ontology/abbreviation	"EDCOM II"
http://dbpedia.org/ontology/academicAdvisor	http://dbpedia.org/resource/Joseph_Erlanger
http://dbpedia.org/ontology/academicDiscipline	http://dbpedia.org/resource/Artificial_intelligence
http://dbpedia.org/ontology/academicDiscipline	http://dbpedia.org/resource/Statistical_learning_theory
http://dbpedia.org/ontology/academicDiscipline	http://dbpedia.org/resource/Remote_sensing
http://dbpedia.org/ontology/academicDiscipline	http://dbpedia.org/resource/Digital_image_processing
http://dbpedia.org/ontology/academicDiscipline	http://dbpedia.org/resource/Physiology
http://dbpedia.org/ontology/academicDiscipline	http://dbpedia.org/resource/Nuclear_chemistry
http://dbpedia.org/ontology/activeYearsEndDate	2023-11-09
http://dbpedia.org/ontology/activeYearsEndDate	2025-01-01
http://dbpedia.org/ontology/activeYearsEndDate	2024-08-06
http://dbpedia.org/ontology/activeYearsEndDate	2010-06-30

http://dbpedia.org/ontology/activeYearsEndDate	2019-06-30
http://dbpedia.org/ontology/activeYearsEndDate	2022-06-30
http://dbpedia.org/ontology/activeYearsEndDate	2025-07-10
http://dbpedia.org/ontology/activeYearsEndDate	1916-09-04
http://dbpedia.org/ontology/activeYearsEndDate	2024-12-12
http://dbpedia.org/ontology/activeYearsEndDate	2025-06-30
http://dbpedia.org/ontology/activeYearsEndDate	1872-01-05
http://dbpedia.org/ontology/activeYearsEndDate	1974-01-07
http://dbpedia.org/ontology/activeYearsEndDate	1993-12-14
http://dbpedia.org/ontology/activeYearsEndDate	2023-08-03
http://dbpedia.org/ontology/activeYearsEndDate	2024-11-19
http://dbpedia.org/ontology/activeYearsEndDate	2004-03-26
http://dbpedia.org/ontology/activeYearsEndDate	2008-04-01
http://dbpedia.org/ontology/activeYearsEndDate	2007-10-01
http://dbpedia.org/ontology/activeYearsEndDate	2020-10-01
http://dbpedia.org/ontology/activeYearsEndDate	2010-12-29
http://dbpedia.org/ontology/activeYearsEndDate	1865-06-26
http://dbpedia.org/ontology/activeYearsEndDate	2005-05-11
http://dbpedia.org/ontology/activeYearsEndDate	2007-06-30
http://dbpedia.org/ontology/activeYearsEndDate	2016-06-30
http://dbpedia.org/ontology/activeYearsEndDate	1981-12-21
http://dbpedia.org/ontology/activeYearsEndDate	2004-11-01
http://dbpedia.org/ontology/activeYearsEndDate	2010-11-09
http://dbpedia.org/ontology/activeYearsEndDate	2012-01-18
http://dbpedia.org/ontology/activeYearsEndDate	2012-04-24
http://dbpedia.org/ontology/activeYearsEndDate	1904-12-31
http://dbpedia.org/ontology/activeYearsEndDate	1916-12-20
http://dbpedia.org/ontology/activeYearsEndDate	2017-05-18

Q3: Excerpt; the full result is too large to include.

- For each of these applicable properties, except for `rdf:type`, which different values do they take globally for all those instances?

```
#RDF base vocabulary namespace
prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
prefix dbo: <http://dbpedia.org/ontology/>
#use of group_concat to combine the values in a column in a single text
#chain
select ?prop (group_concat(distinct ?val; separator=" | ") as ?Values)
where {
    ?politician rdf:type dbo:Politician .
    ?politician ?prop ?val .
    filter(?prop != rdf:type)
}
group by ?prop
```

prop	numValues
http://dbpedia.org/ontology/wikiPageWikiLink	904887
http://www.w3.org/2002/07/owl#sameAs	800611
http://www.w3.org/2000/01/rdf-schema#label	330100
http://dbpedia.org/ontology/description	264667
http://dbpedia.org/ontology/termPeriod	212440
http://purl.org/dc/terms/subject	148928
http://xmlns.com/foaf/0.1/depiction	127271
http://dbpedia.org/ontology/wikiPageExternalLink	119591

http://dbpedia.org/property/predecessor	110545
http://xmlns.com/foaf/0.1/isPrimaryTopicOf	110427
http://www.w3.org/ns/prov#wasDerivedFrom	110427
http://dbpedia.org/property/successor	103905
http://dbpedia.org/property/name	97453
http://xmlns.com/foaf/0.1/name	97376
http://dbpedia.org/ontology/thumbnail	66091
http://dbpedia.org/property/office	65960

SPARQL | HTML5 table (partial result: anytime timeout)

prop	numValues
http://dbpedia.org/ontology/wikiPageWikiLink	904887
http://www.w3.org/2002/07/owl#sameAs	800611
http://www.w3.org/2000/01/rdf-schema#label	330100
http://dbpedia.org/ontology/description	264667
http://dbpedia.org/ontology/termPeriod	212440
http://purl.org/dc/terms/subject	148928
http://xmlns.com/foaf/0.1/description	127271
http://dbpedia.org/ontology/wikiPageExternalLink	119591
http://dbpedia.org/property/predecessor	110545
http://xmlns.com/foaf/0.1/isPrimaryTopicOf	110427
http://www.w3.org/ns/prov#wasDerivedFrom	110427
http://dbpedia.org/property/successor	103905
http://dbpedia.org/property/name	97453
http://xmlns.com/foaf/0.1/name	97376
http://dbpedia.org/ontology/thumbnail	66091
http://dbpedia.org/property/office	65960
http://dbpedia.org/property/birthPlace	46803
http://dbpedia.org/property/birthDate	45240
http://dbpedia.org/property/spouse	44207
http://dbpedia.org/ontology/birthDate	43588
http://dbpedia.org/property/termStart	37582
http://dbpedia.org/property/termEnd	36520
http://dbpedia.org/property/title	35649
http://dbpedia.org/ontology/birthPlace	34624

Q5: DBpedia returned a partial result (anytime timeout), so the values may be incomplete.